

Logging using SLF4J

1. pom.xml

Create a Maven project and add these dependencies:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.demo</groupId>
  <artifactId>SLF4JDemo</artifactId>
  <version>1.0</version>

  <dependencies>

    <dependency>
      <groupId>org.slf4j</groupId>
      <artifactId>slf4j-api</artifactId>
      <version>1.7.30</version>
    </dependency>

    <dependency>
      <groupId>ch.qos.logback</groupId>
      <artifactId>logback-classic</artifactId>
      <version>1.2.3</version>
    </dependency>

  </dependencies>

</project>
```

Exercise 1

Logging Error and Warning Messages

Create:

LoggingExample.java

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
```

```
public class LoggingExample {  
  
    private static final Logger logger =  
        LoggerFactory.getLogger(  
            LoggingExample.class);  
  
    public static void main(String[] args) {  
  
        logger.error(  
            "This is an error message");  
  
        logger.warn(  
            "This is a warning message");  
    }  
}
```

Output

ERROR LoggingExample - This is an error message
WARN LoggingExample - This is a warning message

Exercise 2

Parameterized Logging

Create:

ParameterizedLogging.java

```
import org.slf4j.Logger;  
import org.slf4j.LoggerFactory;  
  
public class ParameterizedLogging {  
  
    private static final Logger logger =  
        LoggerFactory.getLogger(  
            ParameterizedLogging.class);  
  
    public static void main(String[] args) {  
  
        String username = "John";  
  
        int age = 25;
```

```
double salary = 50000;

logger.info(
    "User {} is {} years old",
    username,
    age);

logger.debug(
    "User {} has salary {}",
    username,
    salary);

logger.warn(
    "Warning for user {}",
    username);

logger.error(
    "Error occurred for user {}",
    username);
}
}
```

Output

```
INFO User John is 25 years old
DEBUG User John has salary 50000.0
WARN Warning for user John
ERROR Error occurred for user John
```

Exercise 3

Using Different Appenders

Create a folder:

src/main/resources

Inside it create:

logback.xml

```
<configuration>

    <appender name="console"
        class="ch.qos.logback.core.ConsoleAppender">
```

```
<encoder>
  <pattern>
    %d{HH:mm:ss.SSS}
    [%thread]
    %-5level
    %logger{36}
    - %msg%n
  </pattern>
</encoder>

</appender>

<appender name="file"
  class="ch.qos.logback.core.FileAppender">

  <file>app.log</file>

  <append>true</append>

  <encoder>
    <pattern>
      %d{HH:mm:ss.SSS}
      [%thread]
      %-5level
      %logger{36}
      - %msg%n
    </pattern>
  </encoder>

</appender>

<root level="DEBUG">

  <appender-ref ref="console"/>

  <appender-ref ref="file"/>

</root>

</configuration>
```

Create:

AppenderExample.java

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class AppenderExample {

    private static final Logger logger =
        LoggerFactory.getLogger(
            AppenderExample.class);

    public static void main(String[] args) {

        logger.trace(
            "Trace message");

        logger.debug(
            "Debug message");

        logger.info(
            "Info message");

        logger.warn(
            "Warning message");

        logger.error(
            "Error message");

        System.out.println(
            "Logs written to console and app.log");
    }
}
```

Expected Results

Exercise 1

ERROR - This is an error message

WARN - This is a warning message

Exercise 2

INFO - User John is 25 years old

DEBUG - User John has salary 50000.0

WARN - Warning for user John

ERROR - Error occurred for user John

Exercise 3

- Logs appear in the Eclipse Console.
- A file named:

app.log

PROJECT STRUCTURE







